Q1. Match the following mathematical concepts with the respective analytical challenge they most appropriately address:
1. Conic Sections: Ellipse Characteristic
I. Analyze the orientation and eccentricity based on a given equation.
II. Determine the locus using equation parameters.
2. Three-Dimensional Geometry Characteristic
III. Find angle between two lines given their direction ratios.
IV. Calculate distance between two skew lines in space.
3. Units and Measurements Characteristic
V. Determine the error in measurement using significant figures and least count.
VI. Apply dimensional analysis to check the consistency of a physical equation.
A. I-VI, II-V, III-IV
B. I-IV, II-VI, III-II
C. I-V, II-IV, III-VI
D. I-II, II-III, III-IV

Answer: Option 4
Solution: For Conic Sections: Ellipse, calculating locus using equation parameters fits well. In Three-Dimensional Geometry, calculating the distance between two skew lines requires understanding of coordinates and geometry in space. For Units and Measurements, dimensional analysis is directly applied to physical equations. Hence, Option (4) is the right answer.

Q2. Match the following sections in mathematics with the appropriate analytical approach:
1. Conic Sections: Ellipse Characteristic
I. Deduce properties like foci and directrix from the standard form.
II. Assess the impact of altering the major and minor axes on the shape.
2. Three-Dimensional Geometry Characteristic
III. Evaluate the shortest route a line can take between two points not lying on the same plane.
IV. Formulate the equation of a plane through three non-collinear points.
3. Units and Measurements Characteristic
V. Analyze the precision of a derived unit using base units and dimensional analysis.
VI. Calculate the possible error in a complex derived unit.
A. I-V, II-VI, III-IV
B. I-III, II-IV, III-VI
C. I-VI, II-IV, III-II
D. I-II, II-III, III-V

Answer: Option 1
Solution: In Conic Sections: Ellipse, deducing properties like foci and directrix from the standard form is a characteristic analytical approach. In Three-Dimensional Geometry, formulating the equation of a plane through three non-collinear points is a complex spatial problem. In Units and Measurements, analyzing the precision of a derived unit using base units and dimensional analysis fits the analytical requirement. Hence, Option (1) is the right answer.

Q3. Match the following mathematical concepts with their corresponding analytical challenges:
1. Conic Sections: Ellipse Characteristic
I. Investigate the relationship between the geometric properties and the coefficients of its equation.
II. Calculate area enclosed by ellipse given its axes.
2. Three-Dimensional Geometry Characteristic
III. Analyze the orientation and intersection of lines in space given their equations.
IV. Determine properties of parallelism and perpendicularity in skew lines.

3. Units and Measurements Characteristic
V. Use dimensional analysis for verifying the formula of kinetic energy.
VI. Establish measurement uncertainty principles in experimental physics.

A. I-VI, II-V, III-IV
B. I-IV, II-VI, III-II
C. I-V, II-III, III-VI
D. I-II, II-III, III-IV

Answer: Option 1
Solution: For Conic Sections: Ellipse, investigating the relationship between geometric properties and the coefficients of its equation is analytically challenging. For Three-Dimensional Geometry, determining properties like parallelism and perpendicularity in skew lines requires deep understanding of spatial geometry. In Units and Measurements, using dimensional analysis to verify the formula of kinetic energy involves critical thinking on units and dimensions. Hence, Option (1) is the right answer.